

Standard Operating Procedures: Manufacturing & Quality Assurance

Mandatory Operational Limits, Quality Control Points (QCP), Critical Control Points (CCP) & Deviation Protocols

Scope: Dairy Powder Processing, Nutritional Blending, and High-Speed Packaging Operations

Document ID:	MFG-QA-SOP-2026-v4.2	Effective Date:	01 January 2026
Audit Standard:	FSSC 22000 / ISO 9001 / International Food Safety	Review Cycle:	Annual (Controlled Document)
Applicability:	Processing Silos, Dryers, Filling & Secondary Packaging	Classification:	Confidential — Standard Operating Procedure

1. SOP-QA-101: Powder Moisture, Physical & Compositional Quality Specifications

Parameter	Whole Milk Powder	Skim Milk Powder	Infant Formula Base	Test Standard
Moisture Content (%)	≤ 3.50%	≤ 3.80%	≤ 3.00%	ISO 5537 / IDF 26
Fat Content (%)	26.0 – 28.5%	≤ 1.25%	Formulation Spec	ISO 1736
Solubility Index (ml)	≤ 0.50 ml	≤ 0.50 ml	≤ 0.20 ml	ISO 8156
Scorched Particles	Disc A (Max)	Disc A (Max)	Disc A (Max)	ADPI Standard

- Moisture Out-of-Specification Protocol & Quarantine Rules:
- Level 1 Deviation (WMP 3.51% – 3.70% | SMP 3.81% – 4.00%):** Quarantine all pallets produced during preceding **60 minutes**. Perform composite re-testing (3× independent samples from top, middle, bottom). Blending (max 10% inclusion rate) permitted *only* with written approval from **Plant Quality Manager**.
 - Level 2 Deviation (WMP > 3.70% | SMP > 4.00%):** Immediate packaging line shutdown. 100% quarantine under ERP status REJECT-MOISTURE. Re-drying or commercial downgrade to animal feed only. **Blending into consumer packs is strictly prohibited**. Mandatory sign-off: **Site Operations Manager + Head of Quality**.

2. SOP-PKG-204: Heat Seal Integrity & Temperature Deviation Protocol

Operational Limits: Target: 175°C±2°C | Acceptable Window: 170°C to 180°C | QCP Alarm: 165°C–169°C

Critical CCP Failure: < 165°C or > 185°C.

CRITICAL CONTROL POINT (CCP) QUARANTINE BOUNDARY RULE:

When a Critical CCP or Major QCP temperature drop occurs:

- Quarantine Scope:** 100% of all packs produced during the duration of the excursion **PLUS a mandatory 15-minute buffer** preceding the logged excursion start.
- Physical & ERP Containment:** Apply Red Quarantine Tape and lock batch status in ERP as HOLD-SEAL-TEMP.
- Destructive Testing Protocol:** Draw $n = 20$ sample packs per affected pallet. Perform vacuum bubble leak test at -50 kPa for 60 seconds (ASTM D3078) and tensile seal strength test (min pass: ≥ 25.0 N/15mm).
- Disposition Authority:** If all 20 samples pass, release under concession signed by **Shift QA Lead**. If even 1 sample fails, the entire batch must be un-bagged, sieved, and repacked.

3. SOP-SAN-302: Clean-in-Place (CIP) Validation & Abort Protocols

Cleaning Phase	Chemical Concentration	Temperature	Minimum Contact Time / Spec
Caustic Wash (NaOH)	1.50% – 2.00%	75°C – 80°C	Minimum 30 minutes continuous circulation
Acid Wash (HNO ₃)	0.80% – 1.20%	65°C – 70°C	Minimum 20 minutes continuous circulation
Final Potable Water Rinse	Freshwater Flush	Ambient	Conductivity < 50 μ S/cm

- CIP Abort & Re-Wash Criteria: If final rinse conductivity > 50 μ S/cm at the end of the scheduled cycle:
- Automated Interlock:** Processing and filling startup is mechanically and digitally interlocked.
 - Mandatory Action:** Extend freshwater rinse for min 15 additional minutes until conductivity ≤ 50 μ S/cm is maintained continuously for 5 minutes. Conduct phenolphthalein swab test at lowest drainage point before line release.

4. SOP-HACCP-405: Foreign Matter & Metal Detection Control

Challenge Testing Standards: Challenge tests must be conducted **every 2 hours**, at shift handover, and after any power restart.

- Certified Test Pieces:** Ferrous (Fe): 1.5 mm | Non-Ferrous (Non-Fe): 2.0 mm | Stainless Steel (SS-316): 2.5 mm.

- **Action upon Challenge Test Failure:** Stop line immediately. Quarantine **100% of production back to the last successful challenge test** (maximum 2 hours). Recalibrate unit, verify with 3 consecutive test cycles, and pass 100% of quarantined stock through validated secondary detector.

5. SOP-GOV-501: Non-Conformance Escalation & Sign-Off Matrix

Incident Tier	Incident Description / Criteria	Immediate Containment	Mandatory Final Sign-off
Tier 1 (Minor)	Aesthetic carton defect, net weight offset within MAV limits	Shift QA Physical Hold Tag	Shift QA Lead
Tier 2 (Major)	QCP temperature drift (165 – 169°C), Level 1 moisture (3.51 – 3.70%)	ERP Batch Status Lock	Plant Quality Manager
Tier 3 (Critical)	CCP failure (Seal < 165°C), Metal detector fail, Level 2 moisture (> 3.70%)	Immediate Line Halt + Red Lock	Site Operations Manager + Head of Quality